

Assignment 1

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1 Problem 1

1(a) Prediction from Sample Moments

The sample contains $n = 1000$ observations. Before fitting any distribution, I first calculated the first four sample moments to describe the location, dispersion, asymmetry, and tail behavior of the sample.

The sample mean is

$$\hat{\mu} = \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i,$$

which gives

$$\hat{\mu} = 0.001104.$$

The sample variance is estimated using the $n - 1$ denominator,

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2,$$

which gives

$$\hat{\sigma}^2 = 0.000096.$$

Skewness measures the asymmetry of the distribution. In standardized form, it is based on the third central moment,

$$\text{Skewness} = \frac{\mu_3}{\sigma^3}.$$

Using the finite-sample bias-corrected estimator, the sample skewness is

$$\widehat{\text{Skewness}} = -0.669837.$$

Because this value is negative, the sample is negatively skewed: the distribution has greater extension toward the left tail than toward the right tail.

Kurtosis measures tail heaviness. I report excess kurtosis, defined as

$$\text{Excess Kurtosis} = \frac{\mu_4}{\sigma^4} - 3.$$

The subtraction of 3 makes the excess kurtosis of a Normal distribution equal to zero. Using the finite-sample bias-corrected estimator gives

$$\widehat{\text{Excess Kurtosis}} = 2.357614.$$

Since the excess kurtosis is substantially greater than zero, the sample has heavier tails than a Normal distribution.

Therefore, the two main features of the sample shape are negative skewness and positive excess kurtosis. I use these two moments to evaluate the candidate distributions before fitting any model.

A Normal distribution is ruled out. A Normal distribution has zero skewness and zero excess kurtosis for every choice of its mean and variance. The sample has both negative skewness and positive excess kurtosis, so changing the Normal parameters cannot reproduce either of these features.

A Lognormal distribution is also ruled out by the skewness. A non-degenerate Lognormal distribution must have positive skewness, whereas the observed sample skewness is negative.

A Student's t distribution can produce positive excess kurtosis and therefore can represent heavier tails than the Normal. However, the Student's t distribution is symmetric and has zero skewness whenever the skewness exists. It therefore cannot reproduce the observed negative skewness.

A negatively skewed Normal Inverse Gaussian (NIG) distribution is plausible. The NIG family allows nonzero skewness, including negative skewness, while also allowing positive excess kurtosis. It can therefore represent both main shape characteristics observed in this sample.

1(b) Fitting the Normal and Examining the 1% Quantile

Although the sample moments suggest that the Normal distribution does not fully describe the data, I next fit a Normal distribution as required by matching the sample mean and variance.

For a Normal random variable,

$$X \sim N(\mu, \sigma^2).$$

Matching the parameters to the sample moments gives

$$\hat{\mu} = 0.001104$$

and

$$\hat{\sigma}^2 = 0.000096.$$

Using the unrounded sample variance, the corresponding fitted standard deviation is

$$\hat{\sigma} = \sqrt{\hat{\sigma}^2} = 0.009811.$$

Thus, the fitted model is approximately

$$X \sim N(0.001104, 0.000096).$$

For a Normal distribution, a quantile can be obtained from the standard Normal quantile. The p -quantile is

$$q_p = \hat{\mu} + \hat{\sigma}\Phi^{-1}(p),$$

where $\Phi^{-1}(p)$ is the standard Normal quantile function.

For $p = 0.01$,

$$q_{0.01} = \hat{\mu} + \hat{\sigma}\Phi^{-1}(0.01).$$

Since

$$\Phi^{-1}(0.01) \approx -2.326,$$

the fitted Normal's 1% quantile is

$$q_{0.01} = 0.001104 + 0.009811(-2.326) = -0.021719.$$

By definition, the fitted Normal assigns 1% probability to observations below this value. With $n = 1000$ observations, the expected number below the 1% quantile is therefore

$$n(0.01) = 1000(0.01) = 10.$$

The code then counts the number of actual observations satisfying

$$x_i < -0.021719.$$

There are 26 such observations in the sample, compared with only 10 expected under the fitted Normal.

1(c) Reconciliation

The fitted Normal substantially underestimates the frequency of extreme negative observations.

Under the fitted Normal model, only 1% of observations should fall below

$$-0.021719.$$

However, the observed fraction is

$$\frac{26}{1000} = 0.026 = 2.6\%.$$

Thus, observations in this region of the left tail occur approximately

$$\frac{2.6\%}{1\%} = 2.6$$

times as frequently as the fitted Normal predicts.

The error therefore appears specifically in the left tail: the fitted Normal assigns too little probability to sufficiently large negative outcomes. This result is consistent with the prediction from part (a). The negative sample skewness indicated an asymmetric distribution with greater extension toward the left tail, while the positive excess kurtosis indicated heavier tails than the Normal distribution.

The observed count of 26 observations below the Normal 1% quantile provides direct evidence that the fitted Normal does not assign enough probability to the extreme left tail.

2 Problem 2

2(a) Prediction of the Error Distribution

Before fitting any regression model, I first plot y against x .

The scatter plot shows a strong approximately linear and positive relationship between x and y . Most observations appear to be concentrated around a common straight-line pattern, suggesting that a linear specification of the form

$$y_i = \alpha + \beta x_i + \epsilon_i$$

is reasonable.

For a given x_i , the error term can be interpreted as the vertical deviation of an observation from the underlying linear relationship,

$$\epsilon_i = y_i - (\alpha + \beta x_i).$$

Most observations in Figure 1 appear to have relatively small vertical deviations. However, there are also several observations substantially farther from the main linear pattern than the majority of the sample. These unusually large deviations suggest that extreme errors may occur more frequently than would be expected under a Normal error distribution.

I therefore predict that the error term follows a Student's t distribution rather than a Normal distribution. The Student's t distribution is symmetric around its location, like the Normal distribution, but has fatter tails. This allows the errors to remain approximately symmetric around zero while assigning greater probability to unusually large positive or negative residuals.

Thus, before fitting either model, my prediction is

$$\epsilon_i \sim t_\nu(0, s),$$

for some degrees-of-freedom parameter ν and scale parameter s , rather than a Normal error distribution.

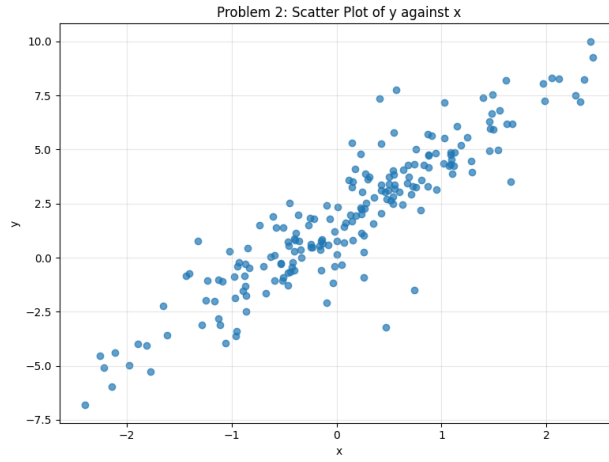


Figure 1: Scatter plot of y against x

2(b) OLS Assumption and Predicted Effects

The apparent violation is Assumption 7: the regression error term is normally distributed. Based on the scatter plot, I expect the errors to be symmetric but more heavy-tailed than a Normal distribution.

The violation by itself may not create a large systematic bias in the estimated slope. The main linear relationship between x and y remains clear. However, because OLS minimizes the sum of squared residuals,

$$\sum_{i=1}^n \hat{\epsilon}_i^2,$$

a small number of unusually large residuals can receive substantial weight and may pull the estimated slope away from the value obtained under a heavy-tailed error model.

For the standard error of the slope, the OLS estimate depends on the estimated residual variance,

$$s^2 = \frac{\sum_{i=1}^n \hat{\epsilon}_i^2}{n - p},$$

and

$$SE(\hat{\beta}) = s\sqrt{\text{diag}((X'X)^{-1})}.$$

Because a heavy-tailed error distribution produces large residuals more frequently, these squared residuals may increase the estimated residual variance and therefore increase the estimated standard error of the slope. Therefore, prediction is made that the effect on the slope itself will be relatively small, while the effect on its estimated uncertainty may be more noticeable.

Model Fitting

The regression model is

$$y_i = \alpha + \beta x_i + \epsilon_i.$$

Three specifications are fitted: OLS, Normal-error maximum likelihood, and Student's t -error maximum likelihood.

Ordinary Least Squares

OLS estimates the coefficients by minimizing

$$\sum_{i=1}^n [y_i - (\alpha + \beta x_i)]^2,$$

giving the matrix solution

$$\hat{\beta}_{\text{OLS}} = (X'X)^{-1}X'Y.$$

The fitted coefficients and their standard errors are

$$\hat{\alpha}_{\text{OLS}} = 1.560740, \quad SE(\hat{\alpha}) = 0.096269,$$

$$\hat{\beta}_{\text{OLS}} = 2.990757, \quad SE(\hat{\beta}) = 0.096640.$$

The residual standard error is $s = 1.351401$, where

$$s^2 = \frac{\sum_{i=1}^n \hat{\epsilon}_i^2}{n-p}, \quad \widehat{\text{Var}}(\hat{\beta}) = s^2(X'X)^{-1}.$$

Thus, the fitted OLS line is

$$\hat{y} = 1.560740 + 2.990757x.$$

Normal-Error Maximum Likelihood

Under the Normal specification,

$$\epsilon_i \sim N(0, \sigma^2),$$

with log-likelihood

$$\ell(\alpha, \beta, \sigma) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n [y_i - (\alpha + \beta x_i)]^2.$$

Maximizing this likelihood with respect to α and β is equivalent to minimizing the OLS sum of squared residuals. Therefore,

$$\hat{\alpha}_N = 1.560740, \quad \hat{\beta}_N = 2.990757.$$

The Normal MLE uses

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n \hat{\epsilon}_i^2,$$

giving

$$\hat{\sigma} = 1.344627.$$

Hence,

$$y_i = 1.560740 + 2.990757x_i + \epsilon_i, \quad \epsilon_i \sim N(0, 1.344627^2).$$

Student's t -Error Maximum Likelihood

For the Student's t specification,

$$\epsilon_i \sim t_\nu(0, s),$$

and α, β, s, ν are estimated jointly by maximizing

$$\ell(\alpha, \beta, s, \nu) = \sum_{i=1}^n \log f_t(y_i - \alpha - \beta x_i; \nu, 0, s).$$

The fitted parameters are

$$\hat{\alpha}_t = 1.547979, \quad \hat{\beta}_t = 3.019179,$$

$$\hat{s} = 0.935509, \quad \hat{\nu} = 3.632166.$$

Thus,

$$y_i = 1.547979 + 3.019179x_i + \epsilon_i, \quad \epsilon_i \sim t_{3.632166}(0, 0.935509).$$

Here s is a scale parameter rather than the standard deviation. For $\nu > 2$,

$$\text{Var}(\epsilon) = s^2 \frac{\nu}{\nu - 2}.$$

The fitted value $\hat{\nu} = 3.632166$ indicates substantial tail heaviness relative to a Normal distribution.

AICc Model Comparison

Model selection uses

$$AIC = 2k - 2 \log(\hat{L}), \quad AIC_c = AIC + \frac{2k^2 + 2k}{n - k - 1}.$$

The Normal model has $k = 3$ parameters (α, β, σ) , while the Student's t model has $k = 4$ (α, β, s, ν) . The fitted results are

$$AIC_{c,N} = 692.144523, \quad AIC_{c,t} = 665.327487,$$

with

$$\Delta AIC_c = 692.144523 - 665.327487 = 26.817036.$$

Since lower AICc is preferred, the Student's t model is strongly favored despite its additional parameter.

2(c) Reconciliation of the Slopes

The fitted slopes are

$$\hat{\beta}_{\text{OLS}} = \hat{\beta}_N = 2.990757, \quad \hat{\beta}_t = 3.019179.$$

The difference between the Normal and Student's t estimates is only

$$3.019179 - 2.990757 = 0.028422,$$

or approximately

$$\frac{0.028422}{2.990757} \times 100\% \approx 0.95\%.$$

Thus, the violation of the Normal-error assumption changes the estimated slope only slightly. This is consistent with the prediction that the strong central linear relationship would remain relatively stable, even though extreme residuals could influence an OLS fit.

2(d) What the Normal Model Gets Wrong

Although the fitted slopes are nearly identical,

$$2.990757 \approx 3.019179,$$

the Student's t model improves AICc by 26.817036. Therefore, the main deficiency of the Normal model is not the estimated slope, but the distribution of the residuals.

The Normal model estimates $\hat{\sigma} = 1.344627$, while the Student's t model combines a smaller central scale,

$$\hat{s} = 0.935509,$$

with heavy tails determined by

$$\hat{\nu} = 3.632166.$$

This allows the Student's t model to describe a pattern in which most errors remain relatively close to the regression line while unusually large positive or negative errors occur more frequently than a Normal distribution would predict. Therefore, the Normal model describes the conditional mean well but misrepresents the tails of the error distribution.

2(e) Error Quantiles and Capital Implications

The fitted upper-tail error quantiles are

	95%	99.5%
Normal	2.211715	3.463530
Student's t	2.053833	4.620084

At the 95% level,

$$q_{0.95}^N = 2.211715 > q_{0.95}^t = 2.053833,$$

so the Normal model is wider. At the 99.5% level, the ordering reverses:

$$q_{0.995}^t = 4.620084 > q_{0.995}^N = 3.463530.$$

The reversal results from the interaction between scale and tail shape. The Normal model has the larger central scale ($\hat{\sigma} = 1.344627$ versus $\hat{s} = 0.935509$), which produces the larger 95% quantile. However, the low Student's t degrees of freedom, $\hat{\nu} = 3.632166$, produces much heavier tails. Farther into the tail, this effect dominates the smaller scale and produces the larger 99.5% quantile.

For a capital buffer against rare extreme losses, the Student's t model is preferable. At the 99.5% level it implies an error magnitude of 4.620084, compared with 3.463530 under the Normal model. Thus, the Normal specification would materially underestimate sufficiently extreme tail risk, consistent with the AICc evidence favoring the Student's t model.

3 Problem 3

Prediction from Pairwise Scatter Plots

Judging from Figure 2, the largest disagreement between Pearson and Spearman correlation is expected for the pair (x_1, x_2) . The relationship is strongly monotonic and increasing, but clearly nonlinear. Spearman correlation is based on ranks and therefore captures a strong monotonic relationship, whereas Pearson correlation measures linear association. Consequently, Spearman correlation is expected to be substantially larger than Pearson correlation for this pair.

The pair (x_2, x_3) may also show some disagreement. Its scatter plot retains an overall increasing tendency, but the relationship is distorted by the nonlinear transformation visible in x_2 , with many observations concentrated near $x_2 = 0$. Thus, the relationship is less linear than that between x_1 and x_3 .

In contrast, (x_1, x_3) shows an approximately linear positive relationship, so Pearson and Spearman correlations are expected to be similar. The pairs involving x_4 show little clear linear or monotonic association, so both measures are expected to be near zero and therefore to agree relatively closely.

Therefore, before computing the correlations, the largest expected disagreement is for (x_1, x_2) , while the smallest disagreement is expected among the approximately linear pair (x_1, x_3) and the pairs involving x_4 . The plots alone do not provide sufficient evidence to identify which of these pairs will have the numerically smallest difference.

Fit

The Pearson correlation matrix is

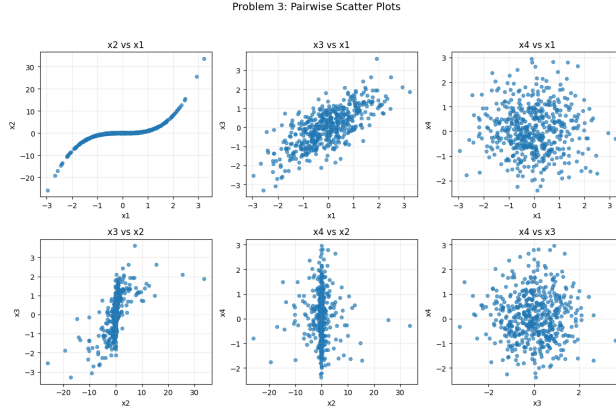


Figure 2: pairwise scatter plot

$$\begin{pmatrix} 1.0000 & 0.7991 & 0.7195 & -0.0047 \\ 0.7991 & 1.0000 & 0.5876 & -0.0209 \\ 0.7195 & 0.5876 & 1.0000 & -0.0086 \\ -0.0047 & -0.0209 & -0.0086 & 1.0000 \end{pmatrix}.$$

The Spearman correlation matrix is

$$\begin{pmatrix} 1.0000 & 0.9719 & 0.6839 & 0.0002 \\ 0.9719 & 1.0000 & 0.6674 & -0.0011 \\ 0.6839 & 0.6674 & 1.0000 & -0.0024 \\ 0.0002 & -0.0011 & -0.0024 & 1.0000 \end{pmatrix}.$$

For each pair, the difference between the two measures is defined as

$$D_{ij} = |\rho_{P,ij} - \rho_{S,ij}|.$$

The largest difference occurs for (x_1, x_2) :

$$\rho_P = 0.799092, \quad \rho_S = 0.971930,$$

$$D_{12} = 0.172838.$$

The smallest difference occurs for (x_1, x_4) :

$$\rho_P = -0.004680, \quad \rho_S = 0.000189,$$

$$D_{14} = 0.004869.$$

Reconcile

The largest difference for (x_1, x_2) agrees with the prediction. For the pair (x_1, x_2) , Spearman correlation is the more honest description of the overall strength of association because the relationship is strongly monotonic but nonlinear. Pearson correlation instead answers the narrower question of how strongly the two variables are linearly related.

The pair (x_2, x_3) also shows a noticeable difference, with Pearson correlation 0.587614 and Spearman correlation 0.667422. This is consistent with its increasing but nonlinear pattern in the scatter plot.

The smallest difference occurs for (x_1, x_4) . Both correlations are approximately zero, indicating that this pair has neither a meaningful linear association nor a meaningful monotonic association. Thus, Pearson and Spearman can agree closely not only when a relationship is linear, but also when essentially no association is present.

4 Problem 4

Predict

The sample covariance matrix is

$$\Sigma = \begin{pmatrix} 1.099875 & 1.696902 \\ 1.696902 & 4.104749 \end{pmatrix}.$$

Here X_1 is the first block and X_2 is the second block, so $\Sigma_{11} = \text{Var}(X_1)$, $\Sigma_{22} = \text{Var}(X_2)$, and $\Sigma_{12} = \Sigma_{21} = \text{Cov}(X_1, X_2)$. Under a multivariate Normal model, the conditional variance of X_2 given X_1 is

$$\text{Var}(X_2 | X_1) = \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}.$$

Substituting the estimated covariance terms,

$$\text{Var}(X_2 | X_1) = 4.104749 - \frac{1.696902^2}{1.099875} = 1.486746.$$

Relative to the unconditional variance of X_2 ,

$$\frac{\text{Var}(X_2 | X_1)}{\text{Var}(X_2)} = \frac{1.486746}{4.104749} = 0.362201.$$

Thus, after observing X_1 , approximately 36.22% of the original variance remains, corresponding to a reduction in uncertainty of about 63.78%.

Under the multivariate Normal model,

$$\text{Var}(X_2 | X_1 = x_1) = \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}.$$

The uncertainty-reduction factor does not depend on the observed value of x_1 . The term

$$\Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}$$

contains only elements of the covariance matrix and contains no x_1 term. Therefore, the conditional variance, and hence the reduction factor, is constant for all observed values of x_1 .

Fit

The conditional expectation under the multivariate Normal model is

$$E[X_2 | X_1 = x_1] = \mu_2 + \Sigma_{21}\Sigma_{11}^{-1}(x_1 - \mu_1).$$

The coefficient on x_1 is

$$\beta = \frac{\Sigma_{21}}{\Sigma_{11}} = \frac{1.696902}{1.099875} = 1.542813.$$

This is also the slope coefficient in the simple linear regression of X_2 on X_1 . Thus, a one-unit increase in X_1 corresponds to an increase of approximately 1.543 units in the conditional expected value of X_2 .

Using the sample means,

$$E[X_2 | X_1 = x_1] = 0.045288 + 1.542813(x_1 + 0.016714).$$

Since the conditional standard deviation is

$$\sigma_{2|1} = \sqrt{1.486746} = 1.219322,$$

the 95% conditional band is

$$E[X_2 | X_1 = x_1] \pm 1.96(1.219322).$$

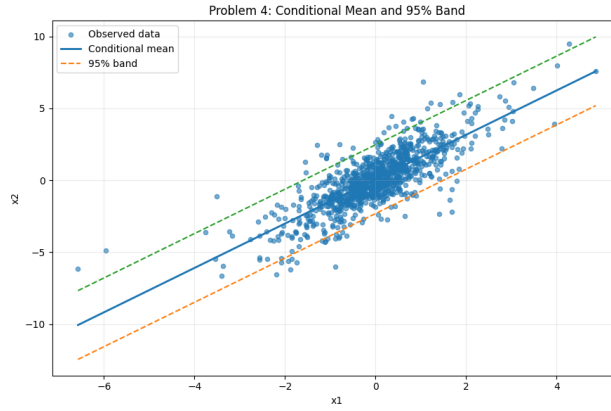


Figure 3: Observed data, conditional expectation, and 95% conditional band.

Overall Coverage

Of the 1000 observations, 935 fall within the 95% conditional band. Therefore, the empirical overall coverage is

$$\frac{935}{1000} = 0.935 = 93.5\%.$$

This is slightly below the 95% coverage predicted by the multivariate Normal model. However, the overall coverage alone does not show whether the model performs equally well across different values of x_1 .

Coverage by Region

The 95% conditional band does not provide uniform coverage across the range of x_1 . The empirical coverage rates are

Region	Coverage
$ x_1 - \bar{x}_1 < 1s_{x_1}$	95.65%
$1s_{x_1} \leq x_1 - \bar{x}_1 < 2s_{x_1}$	85.79%
$ x_1 - \bar{x}_1 \geq 2s_{x_1}$	90.20%

The central region is close to the nominal 95% coverage, while the coverage differs substantially in regions farther from the mean of x_1 . In particular, it falls to 85.79% in the 1–2 standard deviation region and is 90.20% in the most extreme region. Therefore, the overall coverage of 93.5% hides meaningful variation across the range of x_1 .

Reconcile

The empirical coverage is not uniform across the range of x_1 . The central region has coverage close to 95%, while the coverage falls to 85.79% in the 1–2 standard deviation region and 90.20% in the most extreme region.

The failed assumption is the constant-variance assumption. Under the multivariate Normal model,

$$\text{Var}(X_2 | X_1 = x_1) = \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12},$$

which does not depend on x_1 . This implies a constant conditional variance and therefore a constant-width 95% band. The varying empirical coverage indicates that the actual conditional spread of X_2 changes with X_1 . In regression terminology, this is heteroskedasticity.

The conditional mean relation from part (c) can still be useful as a linear description of the center of X_2 given X_1 , and the coefficient 1.542813 can still be interpreted as the slope of that linear relationship. However, the constant conditional variance 1.486746 and the resulting constant-width 95% band are not supported uniformly by the data.

5 Problem 5

Predict

The time-series plot shows fluctuations around a relatively stable level, with no obvious long-term trend or systematic change in variance.

The ACF does not show a sharp cutoff. Instead, it displays an oscillating decay toward zero: the first lag is strongly positive, followed by smaller negative and positive autocorrelations at later lags.

The PACF has two clear significant spikes at lags 1 and 2, with a positive partial autocorrelation at lag 1 and a negative partial autocorrelation at lag 2. Most subsequent lags lie within the 95% significance band.

With $n = 500$, the approximate 95% significance limits are

$$\pm \frac{1.96}{\sqrt{500}} \approx \pm 0.088.$$

Since the ACF decays while the PACF approximately cuts off after lag 2, the pre-fit prediction is an AR(2) model:

$$X_t = c + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t.$$

The plots also suggest that ϕ_1 should be positive and ϕ_2 should be negative.

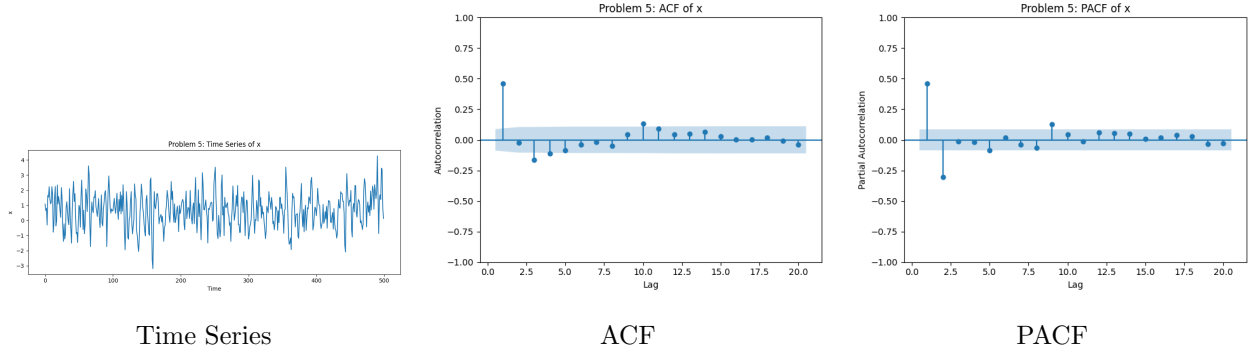


Figure 4: Time series, ACF, and PACF plots

Fit

AR(1)–AR(3) and MA(1)–MA(3) were fitted and compared using

$$AIC_c = AIC + \frac{2k(k+1)}{n-k-1},$$

where k is the number of estimated parameters.

Model	AIC_c
AR(1)	1418.587
AR(2)	1372.438
AR(3)	1374.343
MA(1)	1389.084
MA(2)	1381.232
MA(3)	1374.117

The AR(2) model has the lowest AIC_c , 1372.438, and is therefore selected among the six candidate models.

Its estimated coefficients are

$$\hat{\phi}_1 = 0.601201, \quad \hat{\phi}_2 = -0.302638.$$

Thus, both the selected model and the signs of its coefficients are consistent with the pre-fit prediction from the ACF and PACF.

Reconcile

(d) The AR(2) model has the lowest AICc,

$$AICc_{AR(2)} = 1372.438,$$

compared with 1374.117 for MA(3) and 1374.343 for AR(3). Therefore, AICc selects AR(2).

This agrees with the pre-fit prediction based on the ACF and PACF. The ACF showed a decaying pattern, while the PACF approximately cut off after lag 2, which suggested an AR(2) process.

(e) The AR(2) estimates are

$$\hat{\phi}_1 = 0.601201, \quad \hat{\phi}_2 = -0.302638,$$

while the AR(3) estimates are

$$\hat{\phi}_1 = 0.596203, \quad \hat{\phi}_2 = -0.292690, \quad \hat{\phi}_3 = -0.016542.$$

The third AR coefficient is very small. Adding it improves the log-likelihood only slightly, from

$$-682.178751$$

for AR(2) to

$$-682.110877$$

for AR(3).

AICc trades off goodness of fit against model complexity. Although AR(3) fits the observed sample slightly better, it requires one additional estimated parameter. The small improvement in likelihood is not large enough to compensate for the penalty associated with that extra parameter. As a result,

$$AICc_{AR(2)} = 1372.438 < AICc_{AR(3)} = 1374.343,$$

so the simpler AR(2) model is preferred.

Ordinary R^2 would not make the same tradeoff because it does not penalize the addition of parameters. Adding another regressor cannot decrease R^2 ; it can only increase it or leave it unchanged. Therefore, R^2 could favor the larger AR(3) model even when the third lag contributes almost no useful explanatory power.